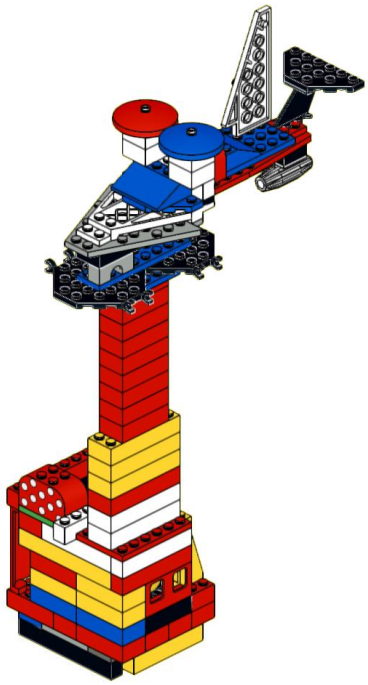
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TECHNICAL COMMUNICATIONS

VERNA JOHNSON



How to Build:

A Base and Control Tower with Landing Pad and Detachable Ship

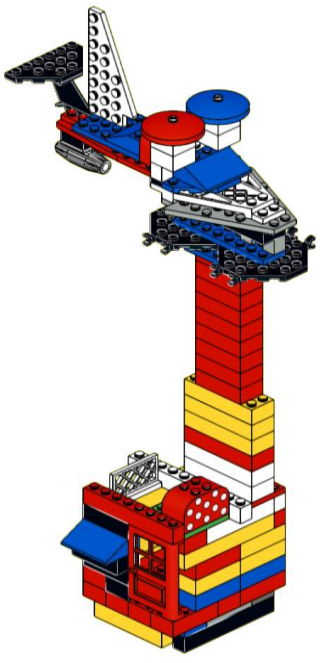


Figure 1 *A completed build of the base, tower with landing pad, and detachable ship.*

GROUP 3

SKYLAR MAGOULIOTIS
MATTHEW LIVINGSTONE
SPENCER MULDER
KYLE YEE

MARCH 20 2026

BEFORE YOU BEGIN



Figure 2 *LEGO is a choking hazard and only recommended for children 3+.*

WARNING: This is an example of a warning

NOTE: This is an example of a note

Figure 3 *An example of a Warning and an example of a Note that you will see throughout the instruction manual.*

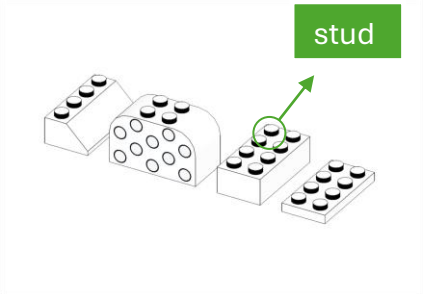


Figure 4 *An example of a 4x3 slope, a 2x4 curved slope, a 2x4 brick, and a 2x4 plate. Every piece has studs.*

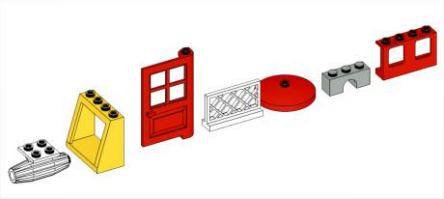


Figure 5 *An example of an engine, windscreen frame, door, fence, radar dish, arch, and window.*



Figure 6 *An example of a wedge plate with grille, wedge plate with cutout, wedge plate with cut corners, modified octagonal plate with clips, wing, and tail.*

How To Build: A Base and Control Tower with Landing Pad and Detachable Ship

Introduction

LEGO is versatile; the user can build creative structures and inspire others. This instruction manual will direct you to build a LEGO structure (Figure 1) that will consist of three major parts: the base (A), a tower with a landing pad (B), and a ship (C). This manual will include instructions on how to build each part of the structure, materials, terms, warnings, and additional notes. Each step will provide you with essential information to build this structure. Warnings will be indicated in yellow, and notes will be indicated in blue (Figure 3). At the end of this manual, you will have a LEGO structure that you can play with, display, or disassemble and re-build into something new. Please ensure you read the “before you begin” section located on the left of page one.

Terms

There are specific terms and dimensions that are used throughout the instruction manual: brick, plate, slope, stud (Figure 4), engine, windscreen frame, door, fence, radar dish, arch, window (Figure 5), wedge plate, modified octagonal plate with clips, wing, and tail (Figure 6).

Materials

LEGO (Figure 7), and instruction manual.
A flat surface is ideal to build any LEGO structure but is not required.

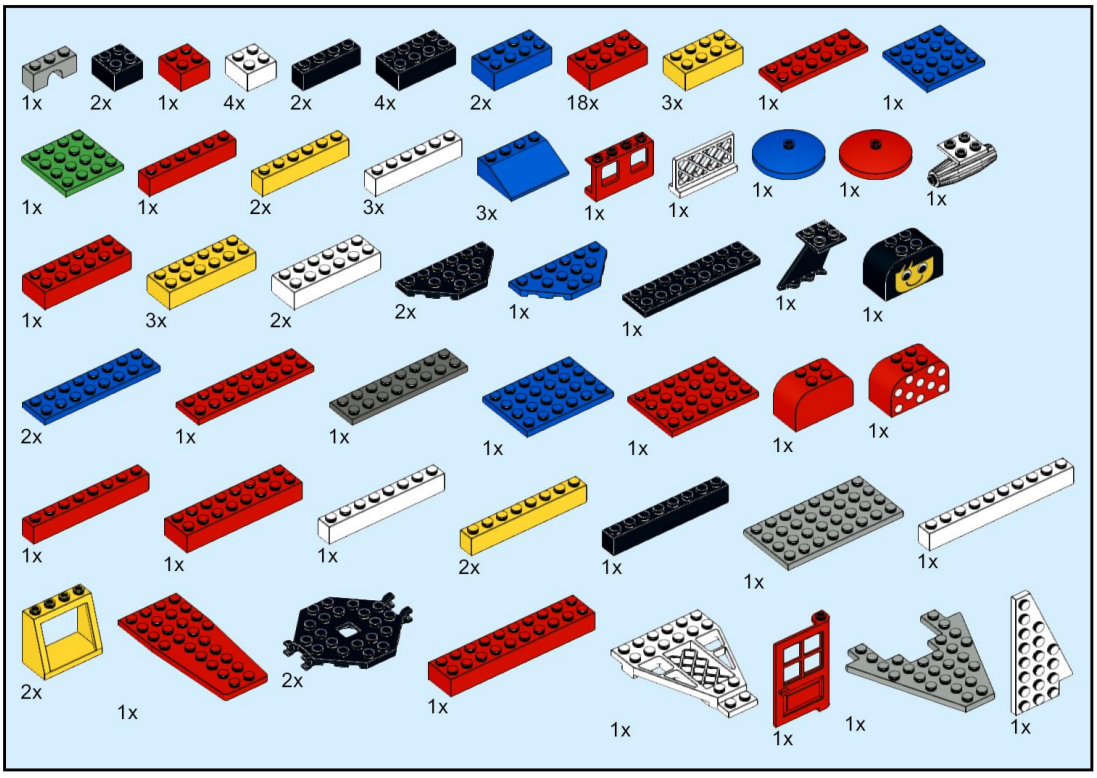


Figure 7 *A list of every piece and the associated quantity that is required to build the base, tower with landing pad, and ship. See figure 1 for an example of the completed build using these pieces. There are 92 pieces required for this build.*

Instructions

First, ensure that you have all the required pieces before you proceed (Figure 7).

Next, build the base.

A) The Base

1. Find one grey 4x8 plate and two (one dark grey and one blue) 2x8 plates.
2. Flip the parts from step 1 upside down and connect them with two (one white and one black)1x8 bricks on each end of the row of plates (Figure 8).
3. Flip what you have built back to the correct orientation (stud-side up).
4. Connect a red 2x8 brick down on the direct center of the 4x8 plate from step 1 so that it lines up lengthwise (Figure 9).

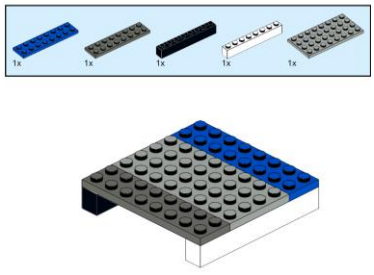


Figure 8

5. Connect four red 2x4 (8 bricks total) on each side of the 2x8 brick (Figure 9).
6. Flip what you have built (stud-side down).

WARNING: Step 7 adds bricks for extra support and stability so that the build does not tip over – if you do not do this step, your base will be unstable, and the structure may fall over while you build.

7. Connect two yellow 1x6 bricks to the sides (connected to the four red 2x6 bricks) (Figure 10).
8. Flip what you have built (stud-side up)
9. Connect a red 2x10 brick (the one with the flames) on one side to act as the foundation of one side wall. It should line up with the base underneath and there should not be any gaps.

WARNING: In the next step, make sure the bricks align in the correct corner so that the door will fit on the base properly (Figure 11).

10. Connect two blue 2x4 bricks in a row, across from the flame piece you placed in the previous step. This will act as the foundation for the other side wall (Figure 11).

NOTE: Step 11 places the windscreen frames. Please ensure that they are oriented so that they slope towards the outside of the base (Figure 12).

11. Connect two yellow windscreen frames to the middle of the red 2x10 brick (with the flames) so that there is one row of studs visible on each end (Figure 12).

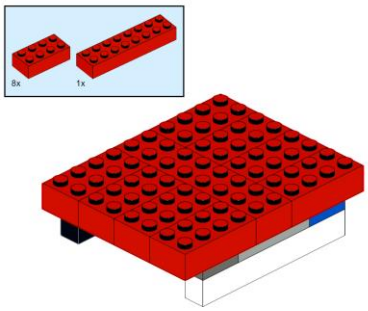


Figure 9

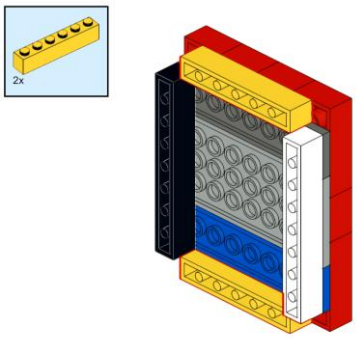


Figure 10

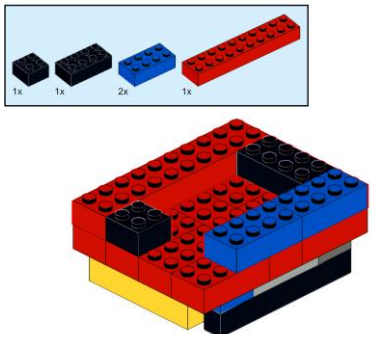


Figure 11

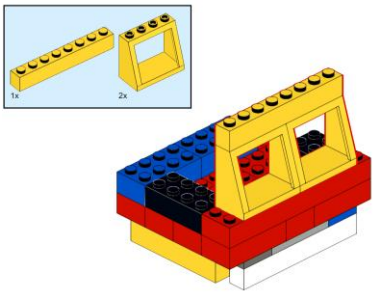


Figure 12

12. Connect one black 2x4 brick to the edge of the base to act as the foundation for the front wall between the two side walls.

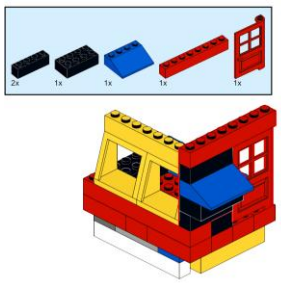


Figure 13

WARNING: Step 13 connects the door to the base. The door might be difficult to work with until it is set in place so be careful. There is a connector on the bottom of the door that must fit in the exact corner stud of the base. If you do not leave a one-stud gap between the door and the two 2x4 bricks from step 10, the door will not open.

NOTE: The door is located at the back of the base.

13. Connect the door to the empty corner stud on the base (See Figure 13 and Figure 14 for correct door orientation).
14. Connect a black 2x2 brick beside the red brick and on the edge of the base to act as a foundation for the back wall.

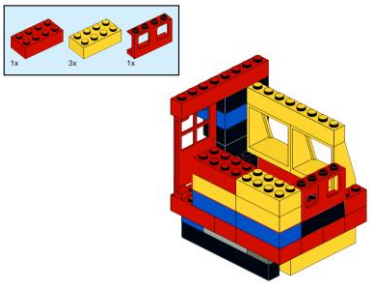


Figure 14

15. Connect a black 1x4 brick on top of the red 2x10 brick (with flames) and the black 2x2 brick placed in the previous step on the edge of the base beside the door (the back wall).
16. Connect a red window piece on top of the black 2x4 brick that is acting as the foundation for the front wall.
17. Build separately (back wall):
    i. Connect a blue slope on top of a black 2x4 brick so that the back of the slope piece is aligned with the black brick piece.
    ii. Connect a black 1x4 brick on top of the blue slope piece from the previous step.
    iii. Connect a red 1x8 brick on top of the step you just built so that it extends to the right.

NOTE: Step 18 will connect the part that will hold the door in place. The part should perfectly align with the size of your base, and you should not have any parts hanging off except for the slope and one brick on the back wall.

18. Connect the part that you built in step 17 on top of the black 1x4 brick that was placed in step 15. Make sure the top of the door is connected under the red 1x8 brick.

19. Build separately (control tower girl):
    i. Connect a black 2x4 brick aligned underneath the curved sloped piece with the face on it.

NOTE: The placement in step 20 is optional. Place the control tower girl in whichever spot you choose.

WARNING: If you want the control tower girl inside of the base, you will need to place her now before the roof is connected.

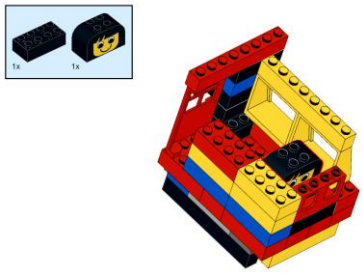


Figure 15

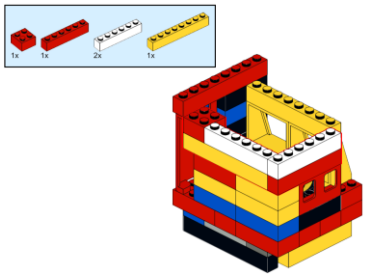


Figure 16

20. Place the piece you built in step 19 in the base so that she is looking out of the red window (Figure 15).
21. Connect a yellow 1x8 brick on top of the yellow windscreen frames (Figure 16).
22. Build separately (parts for side wall):
    i. Connect one yellow 2x4 brick exactly on top of one yellow 2x4 brick.
    ii. Connect one yellow 2x4 brick exactly on top of one red 2x4 brick.
23. Connect the part from step 22.i on top of one blue 2x4 brick acting as the foundation for the side wall to add height to the side wall; Beside this placement, connect the part from step 22.ii on top of one blue 2x4 brick (acting as the foundation of the side wall). This should make the side wall the same height as the front wall (See Figure 14 - Figure 16 for correct placement).
24. Connect a red 1x6 brick on top of the red window and then connect a yellow 1x8 brick on top of the side wall to make a corner and add height to the walls (Figure 16).
25. Connect a white 1x6 brick on top of the red 1x6 brick from the previous step and a white 1x6 brick on top of the yellow 1x6 brick from the previous step to make a corner and add height to the walls (Figure 16).
26. Connect a red 2x2 brick into the available spot beside the white 1x6 brick on the side wall (Figure 16).
27. Connect a green 4x4 plate on top of the red 2x2 brick and part of the white 1x6 brick placed in the previous step. This green plate will act as part of the roof (Figure 17).

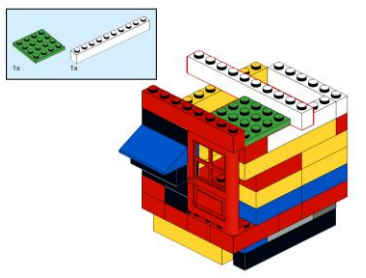


Figure 17

28. Align the white 1x10 brick beside the green plate and connect it on top of the structure so that it is hanging off the yellow window side by two studs and the opposite side by 1 stud (Figure 17).
29. Connect the white fence piece to the wall with the yellow windows directly across from the green plate (Figure 18).
30. Connect the curved slope with ladybug print on top of the outside edge of the green plate (Figure 18).

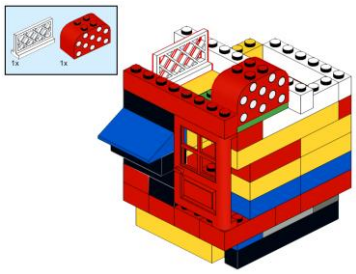


Figure 18

NOTE: The tower (B) and ship (C) will be added to the base after each section is built.

Next, build the tower.

B) The Tower (with landing pad)

1. Take the red 4x6 plate and use it as the base of the tower (Figure 19).
2. Connect six 2x6 bricks together (so that they are vertically aligned) and connect them to the top of the exact center of the plate from the previous step (Figure 20).
3. Connect nine red 2x4 bricks together (so that they are vertically aligned) and then connect them to the top of the exact center of the piece from the previous step (Figure 21).
4. Connect a blue 4x4 plate to the top of the exact center of the piece from the previous step.

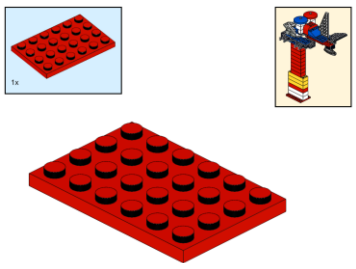


Figure 19

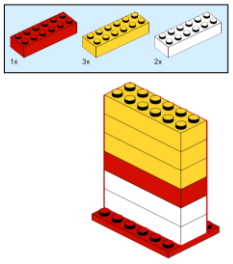


Figure 20

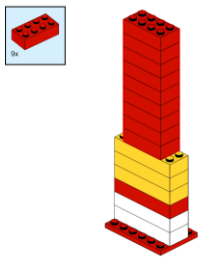


Figure 21

WARNING: Step 5 contains two octagonal plates with clips that must be oriented correctly. The clips must be hanging off the structure from the rectangular plate (Figure 22).

5. Align two modified octagonal plates and connect them by using the blue 2x8 plate on top (Figure 22). This will act as the landing pad.
6. Connect the piece from the previous step to the top of the build in the exact center of the piece from step 4 (Figure 22).

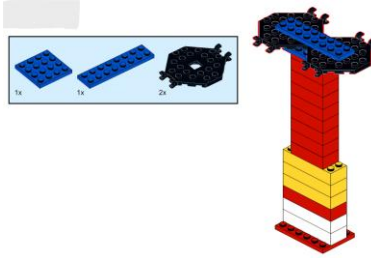


Figure 22

Next, build the ship.

C) The Ship

1. Connect the black 2x2 brick on top of the engine piece (Figure 23).
2. Connect the black 3x6 wedge plate with cut corners on top of the black 4x2x2 tail piece. We will use this later.
3. Connect the red 2x6 plate under the black 2x8 plate so that 4 studs are extending to the right of the black plate. We will use this later.
4. Connect a black 2x4 brick on top of the grey arch piece so that the arch is centered (Figure 23).
5. Connect a grey wedge plate with cutout piece to the top of the piece from the previous step (Figure 24).
6. Connect the white wedge plate with grille piece on top of the piece from the previous step, allowing two of the grey studs to show at the front, and a row of white studs to show at the back (Figure 25).
7. Connect a blue 4x3 slope on top of the blue 3x6 wedge plate with cut corners (Figure 25).
8. Connect the part from step 7 on top of the part from step 6 (Figure 25).

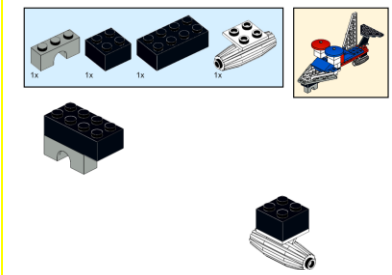


Figure 23

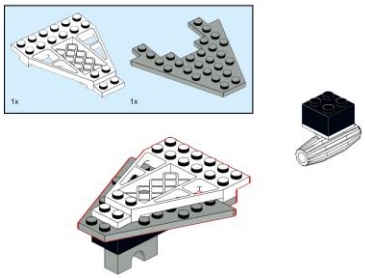


Figure 24

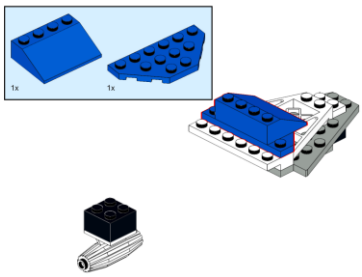


Figure 25

9. Connect the white 1x6 on top of the edge of the black 3x6 wedge plate so that they are aligned (Figure 28).
10. Connect the red curved slope piece to the top (on the available space) of the black wedge plate from the previous step (Figure 28).
11. Connect two black studs on the piece from step 3 underneath the row of studs on from step 6 in the exact center, so that the red plate is farthest away from the front (Figure 26).

WARNING: The red 2x6 plate from step 11 should be at the back of the ship. Please reference the stud placement in Figure 26-28 for correct orientation and placement of pieces for steps 9-14.

12. Connect the red angled piece on top of the piece from step 1. Then connect the red angled piece to the top of the piece from the previous step, so that it is exactly centered (Figure 27).
13. Connect a red 2x8 plate on top of the red angled plate from the previous step so that the back edge is aligned (Figure 28).
14. Connect the piece from step 10 on top so that it is connected to the red angled plate and the white ship piece. The white piece will line up with the blue slope so that they are the same height (Figure 28).
15. Connect a blue 4x3 slope on top of the shorter edge of a blue 4x6 plate so that the flat side of the slope is flush with the plate. Connect this to the red piece from step 13 (Figure 29).

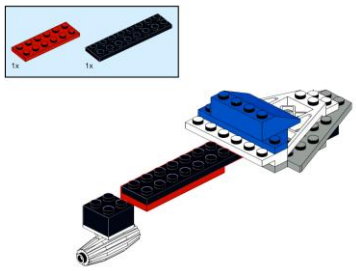


Figure 26

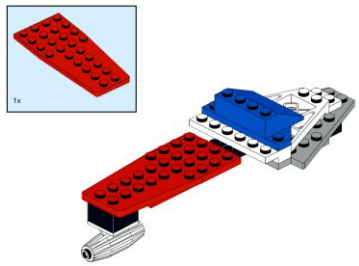


Figure 27

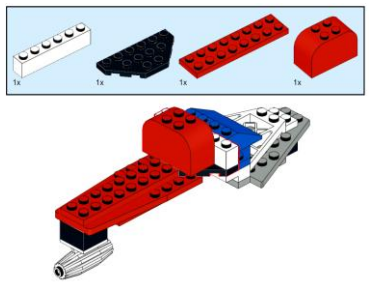


Figure 28

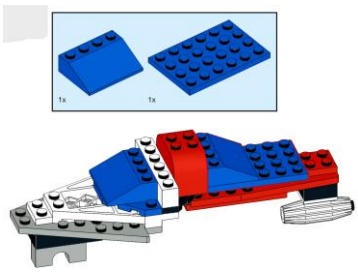


Figure 29

16. Connect the piece from step 2 to the empty space on the red piece from step 13 in the exact center (this will act as the tail of the ship) (Figure 30).
17. Connect two white 2x2 bricks so that they are aligned and connect a red radar dish to the top.
18. Connect two white 2x2 bricks so that they are aligned and connect a blue radar dish to the top.
19. Connect the pieces from step 17 and step 18 to the top of the blue slope and white 1x6 brick from step 14 (Figure 31).

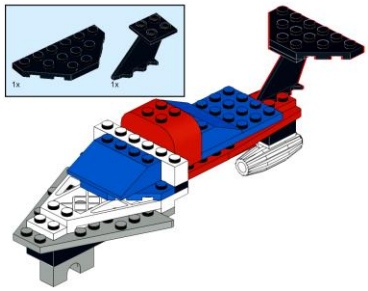


Figure 30

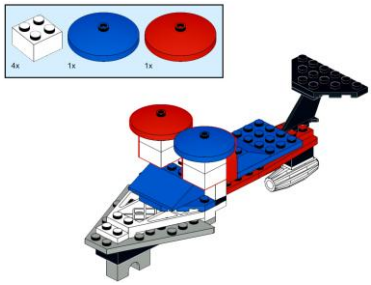


Figure 31

WARNING: The next step is optional. Only attempt if you are an experienced LEGO builder. This is considered an "illegal" placement.

20. Place the white wing piece so that it sticks up on the blue 2x8 plate that is acting as the top of the ship (Figure 32).

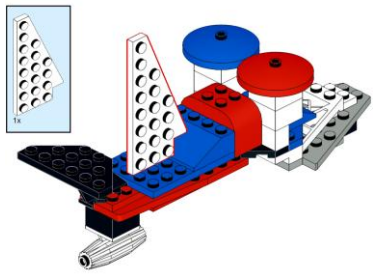


Figure 32

Finally, connect all the parts together (Figure 33).

1. The base of step B will connect to the top of step A, acting as the other half of the roof for the base.
2. The ship is detachable and can be connected to any part of the landing pad from step B.

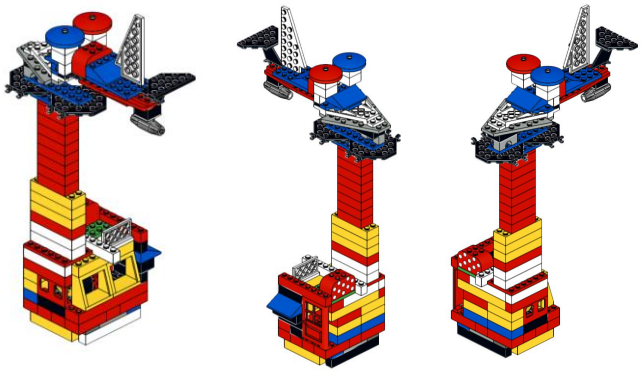


Figure 33 *Three different angles of the completed structure.*

Conclusion

Congratulations on completing the build! By following the instructions correctly, you should have successfully built a base, control tower with a landing pad, and a detachable ship (Figure 33). This completed LEGO structure is something that you can play with, display, or disassemble and re-build into something new.